

Supplementary Figures

Fig. S1 | *tbd* Map of 35 islands surveyed by CREP across the tropical Pacific.

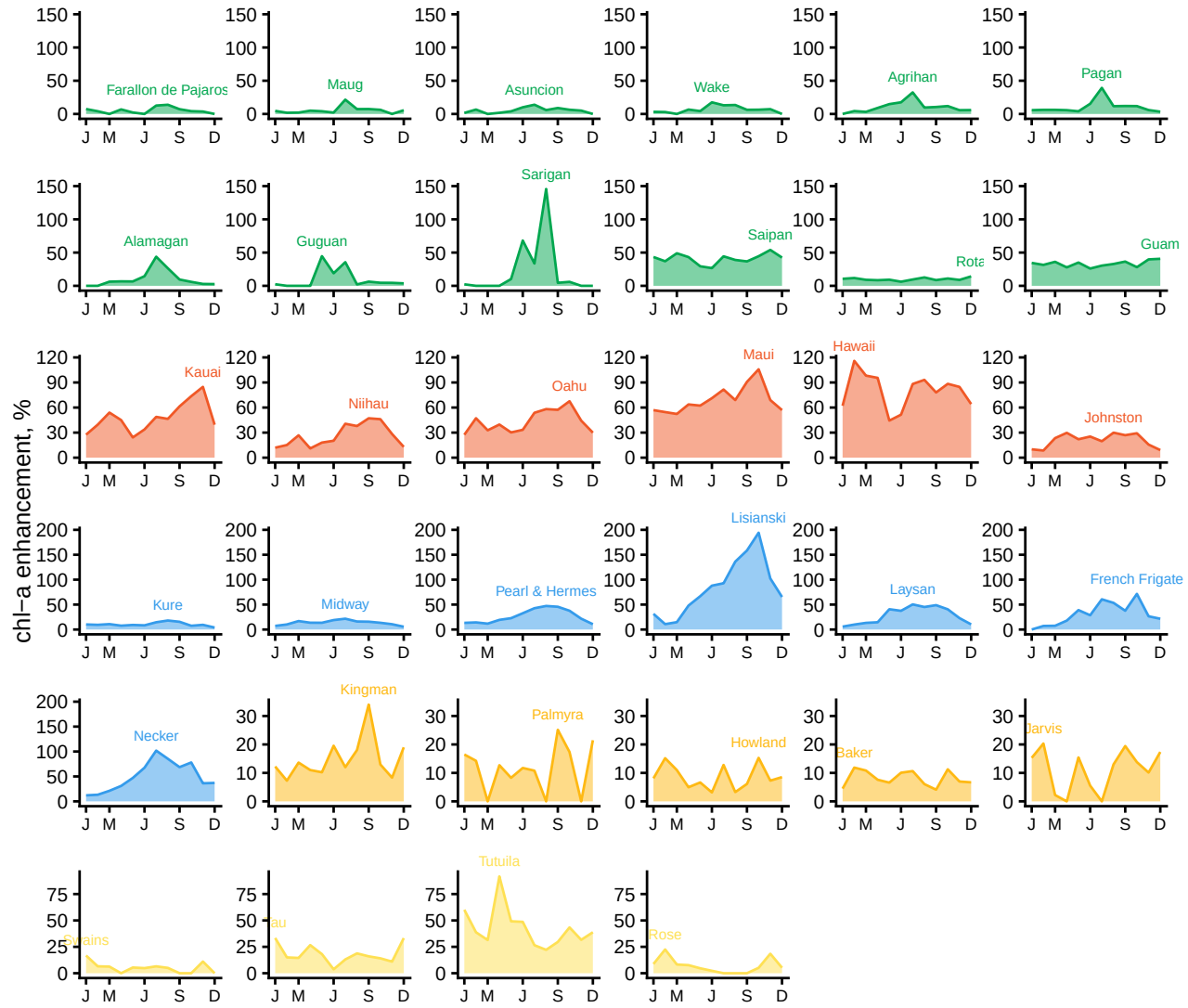


Fig. S2 | Seasonal variation in IME strength at each island. Lines are mean-monthly chl-a enhancement from January to December, ordered and coloured by island group.

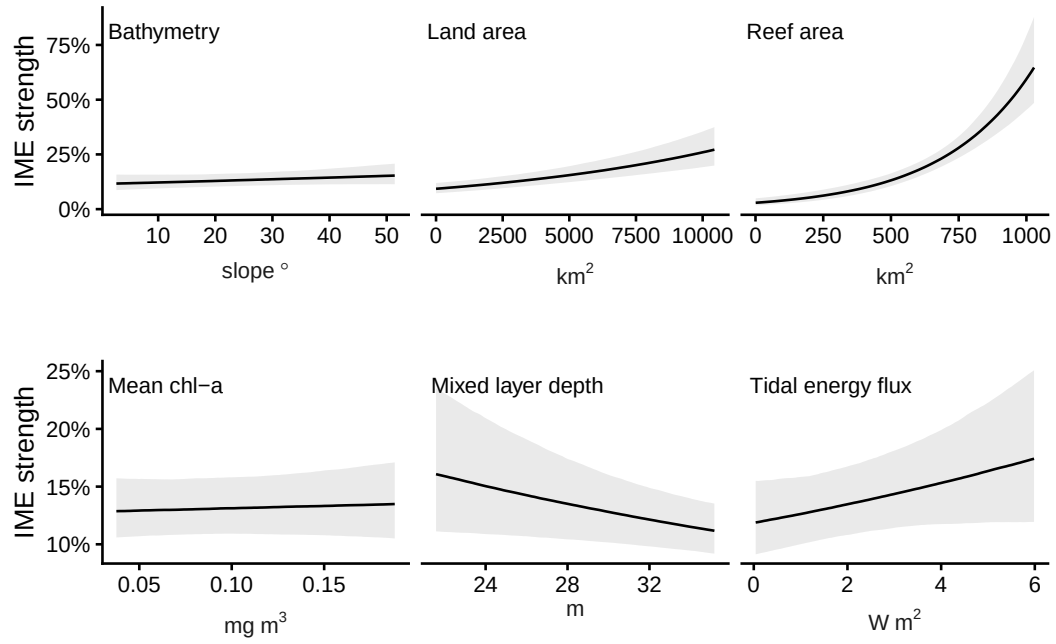


Fig. S3 | Spatial effects on chl-a enhancement for geomorphic features (top) and average oceanographic conditions (bottom) Lines are median posterior predictions ($\pm$ 95% certainty intervals). Note different y-axis values between static and dynamic covariates.

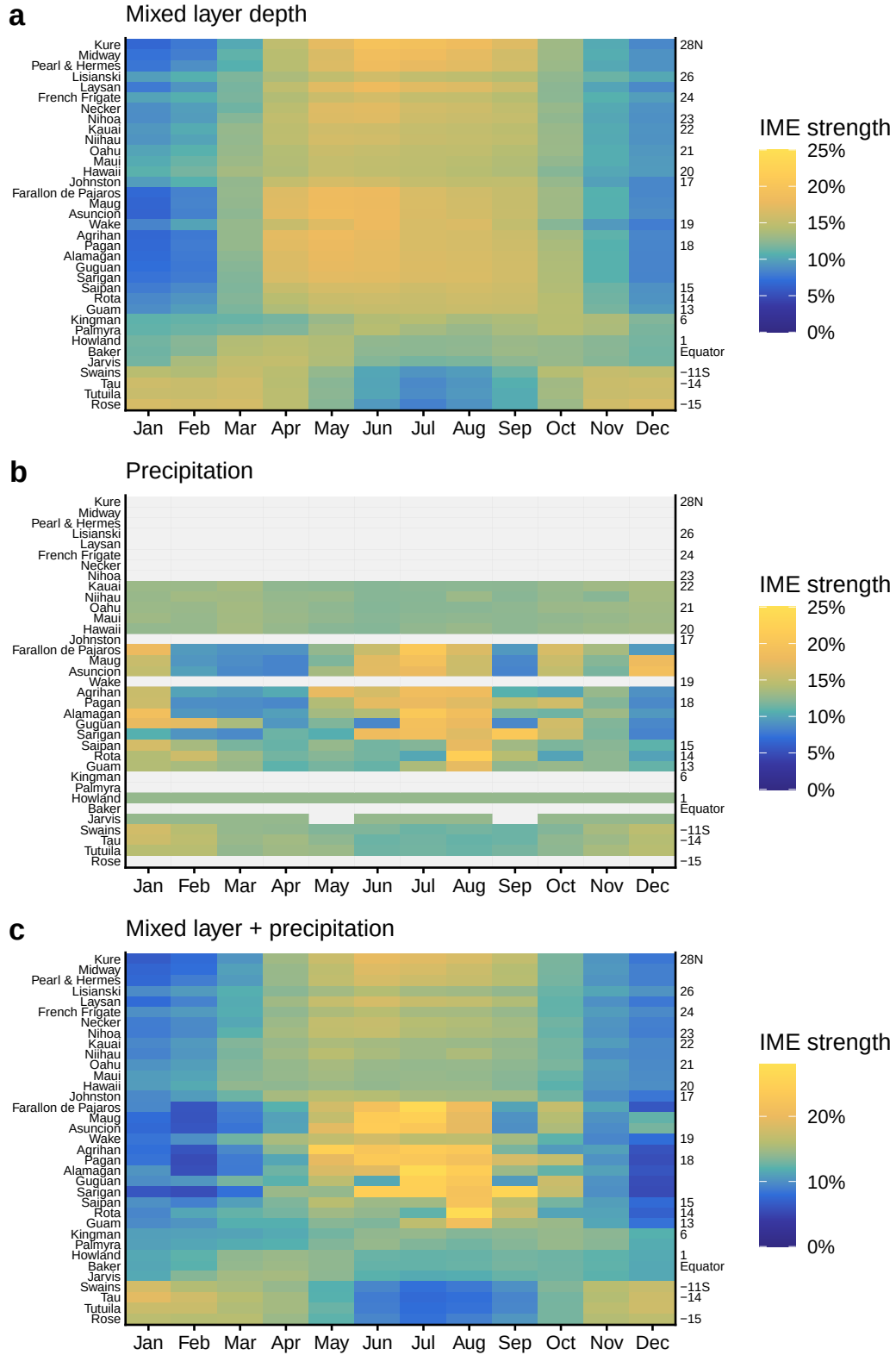


Fig. S4 | Heatmap of seasonal effects on chl-a enhancement for a mixed layer depth, b precipitation and c both covariates combined. Islands with less than 1 mm monthly rainfall are shown in grey. All values are median posterior predictions excluding effects of other covariates.

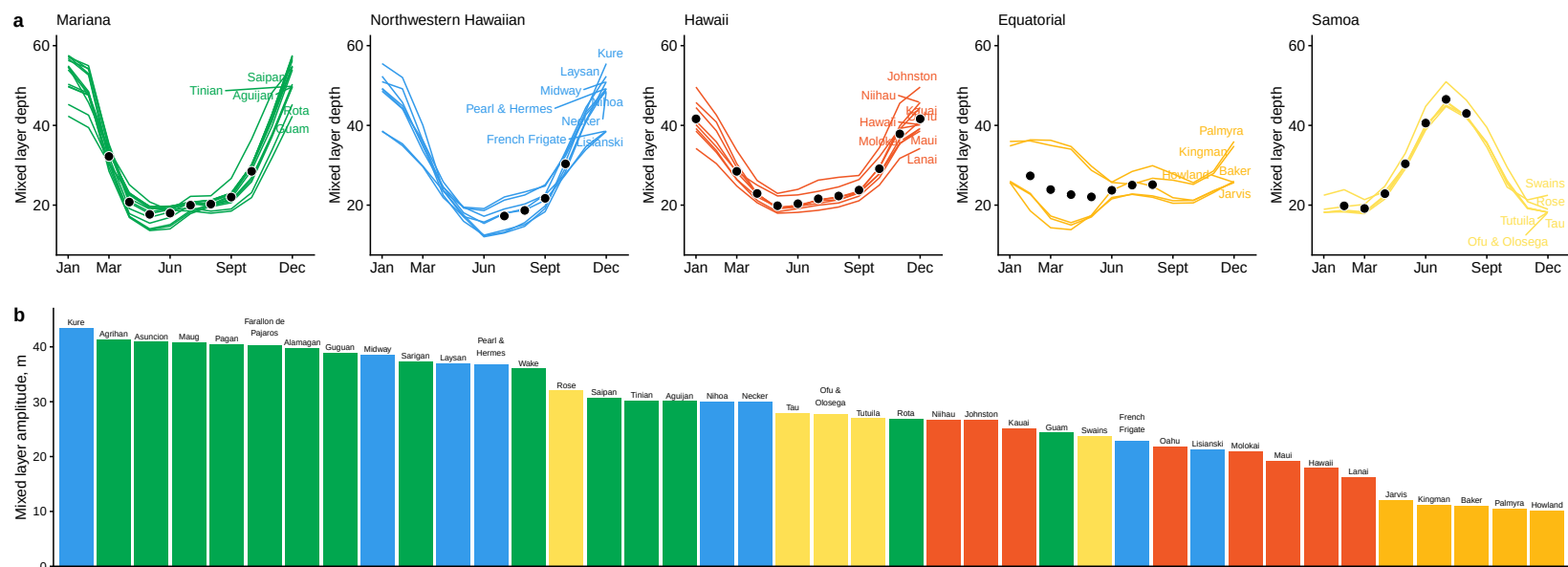


Fig. S5 | Seasonal variability in the mixed layer depth at each island. In a, lines are predicted mixed layer depth at each island from a general additive model of time and month, overlaid with points indicating timing of coral reef fish surveys. Predicted values were used to estimate b the mixed layer depth seasonal amplitude at each island, or the range of depths experienced by each island from Jan to Dec.

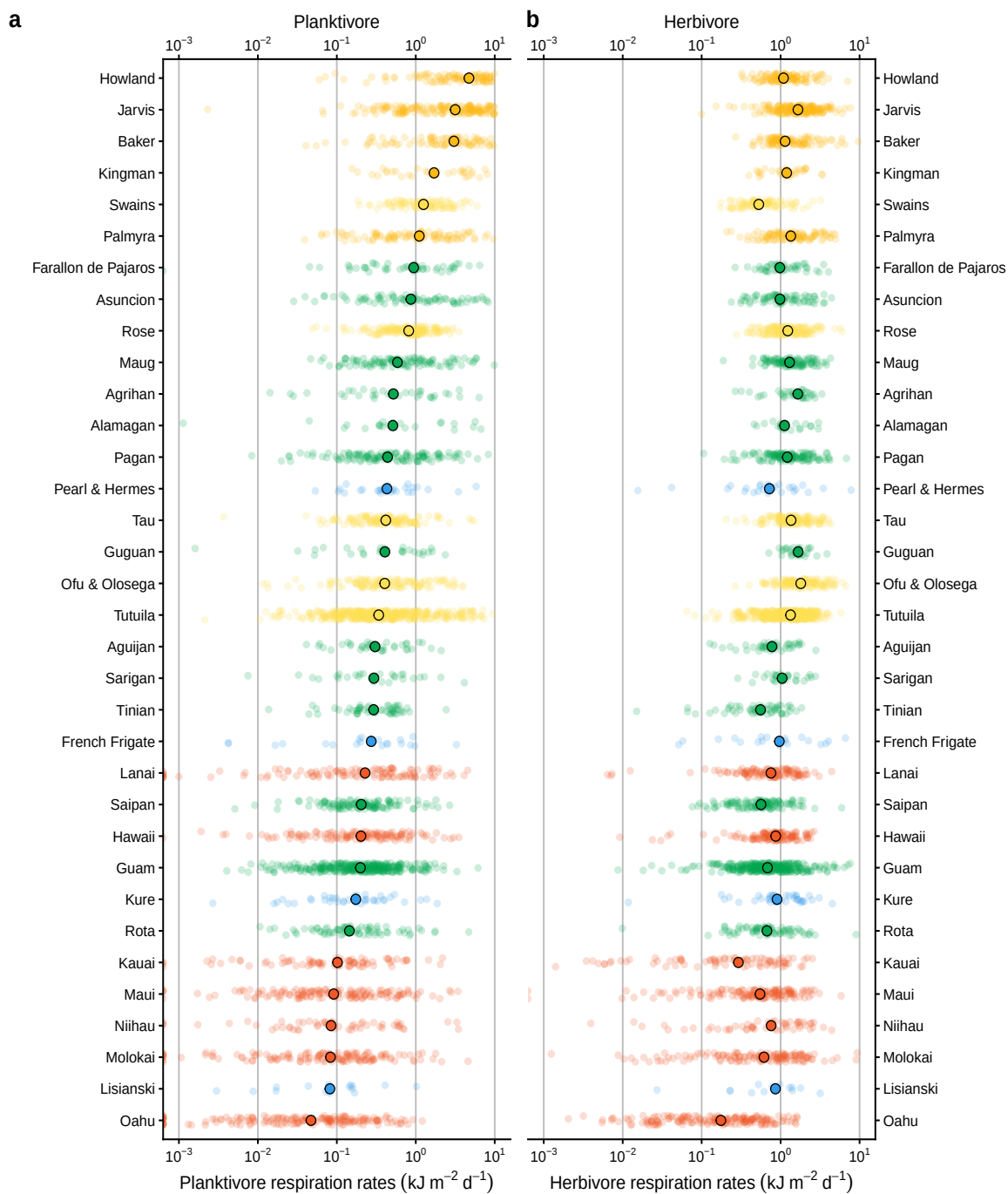


Fig. S6 | Respiration rate of reef-associated planktivores on coral reefs at 34 Pacific islands and atolls. Each point is the median respiration at each island, with jitter showing unique sites ($n = 3,622$).

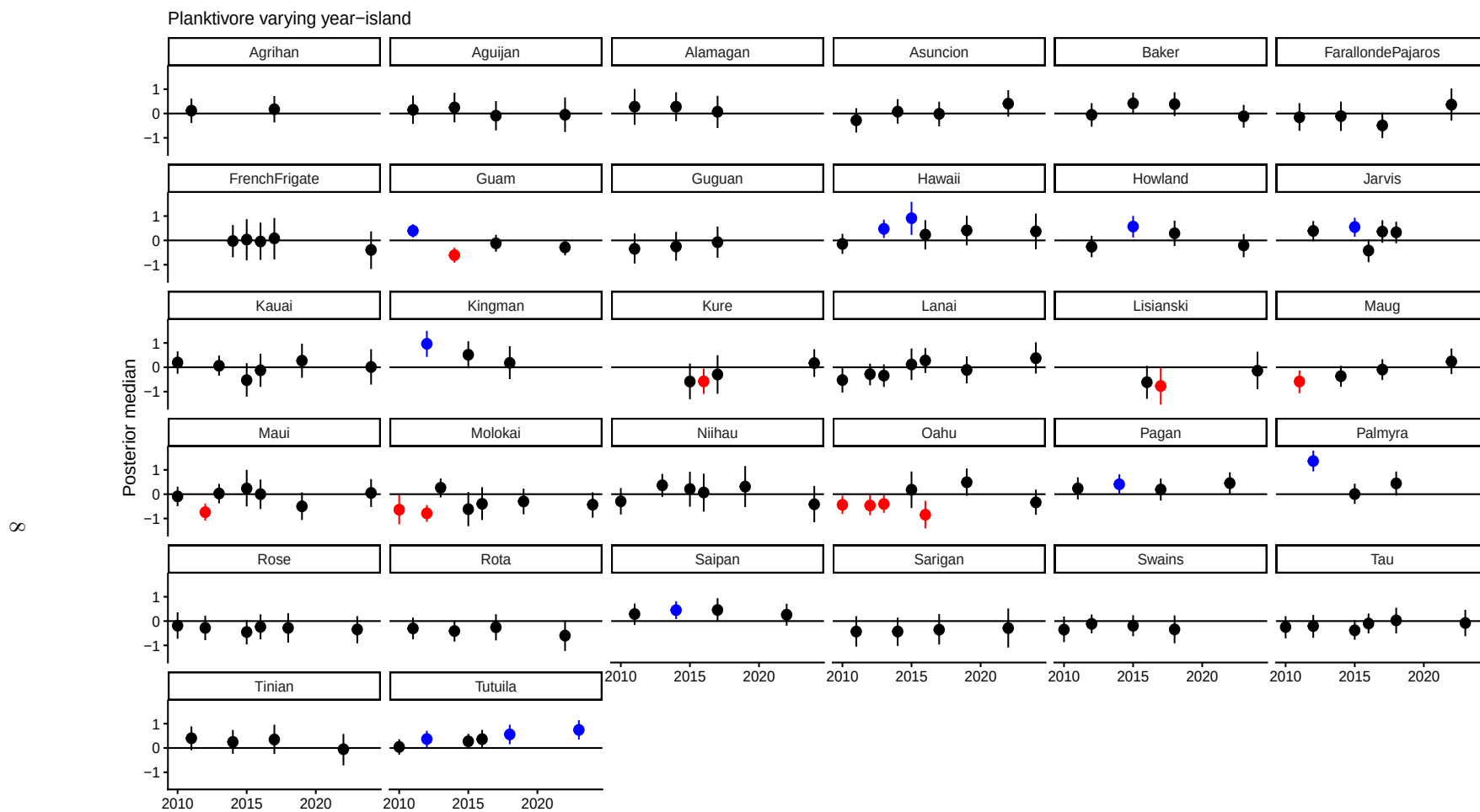


Fig. S7 | Effect of survey year on planktivore respiration rate. Points are posterior medians with 95% certainty intervals, coloured when the posterior did not overlap zero (red = negative, blue = positive).

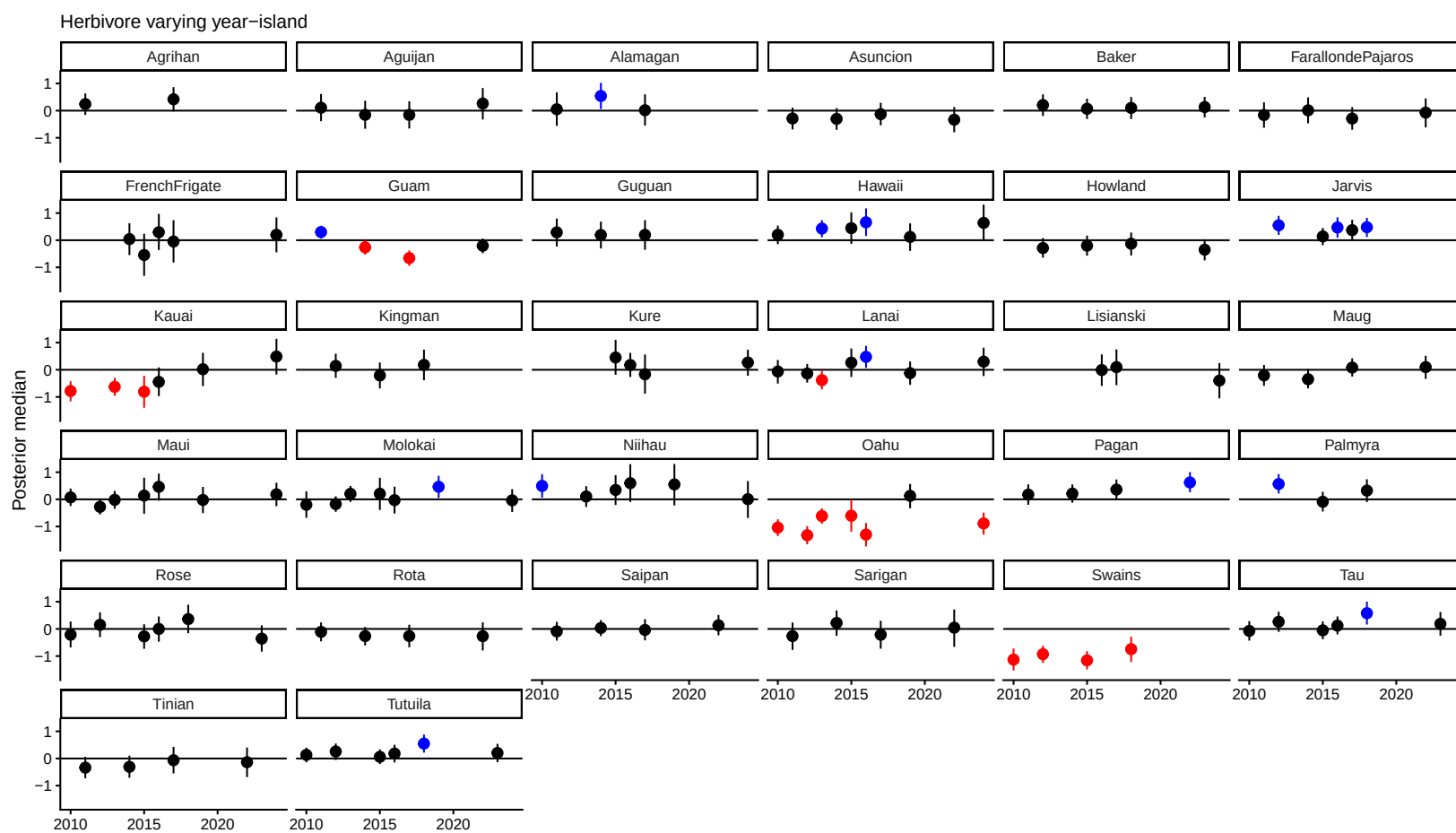


Fig. S8 | Effect of survey year on herbivore respiration rate. Points are posterior medians with 95% certainty intervals, coloured when the posterior did not overlap zero (red = negative, blue = positive).

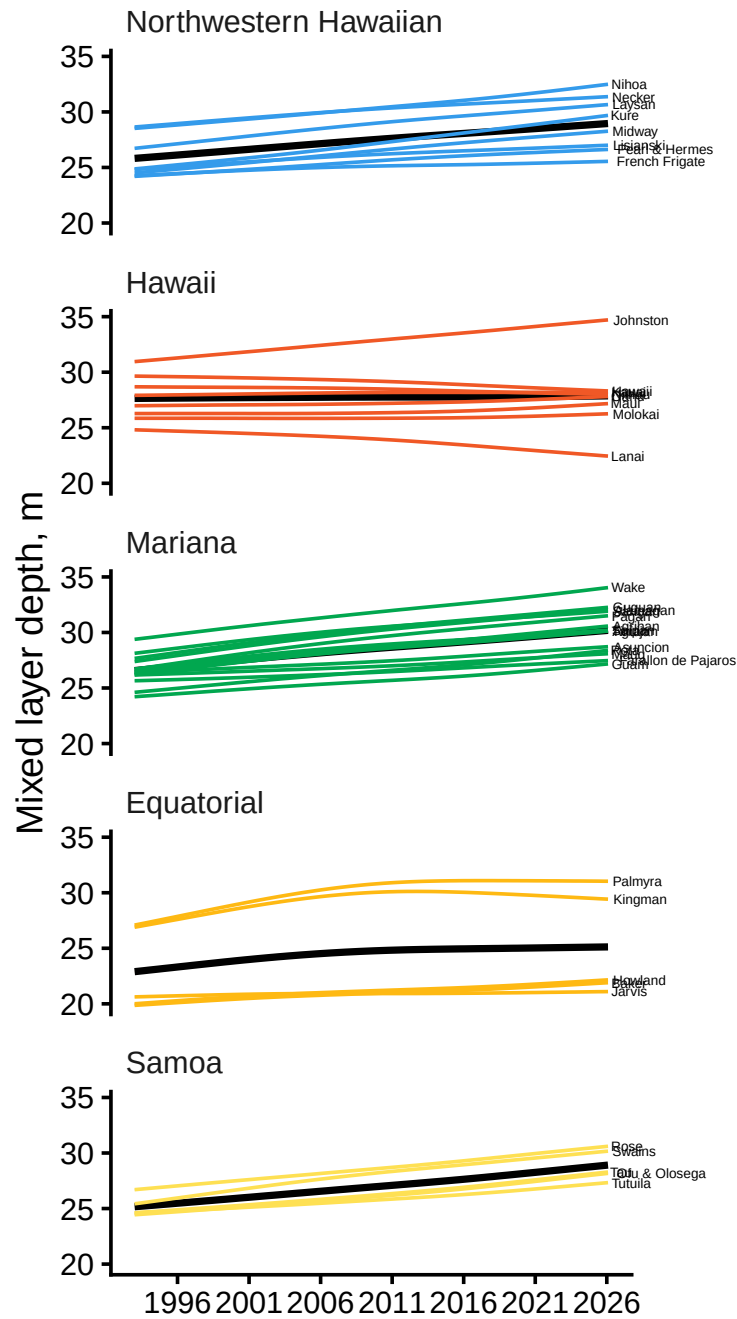


Fig. S9 | Island-level temporal trends in the mixed layer depth layer from 1997-2026. Lines are mean predicted mixed layer from a generalised additive model, for each region (black with 95% CI filled by region), overlaid with mean island predictions (coloured lines).